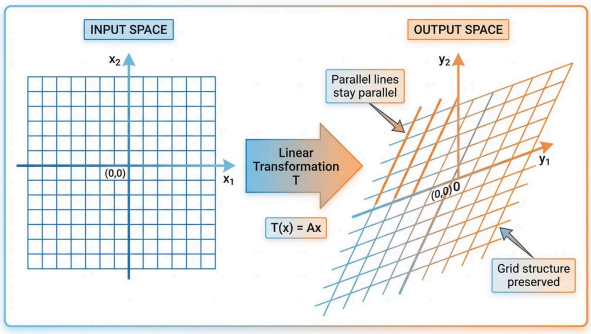
Habib University
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Linear Algebra

MATH 205 L2
Fall Semester 2026

Linear Algebra — Seeing structure where others see numbers.



Course Information

Start Date	End Date	Class Location	Meeting Time
Aug 19 2026	Dec 08 2026	GF-E121	Tues Thurs (09:55 AM-11:10 AM)

Course Prerequisites: MATH 102
Hardware/Software Prerequisites (if any): Computer (Desktop PC or Laptop). Zoom
Content Area: Meets requirements for EE, CS majors, and MATH minor.

Instructor Information

Instructor: Sajal Sohail Rana
Office Location: E-109
Email: sajal.rana@sse.habib.edu.pk
Office Hours: TBT

Course Description

Linear algebra is the study of systems of linear equations. These are ubiquitous in modern sciences and its applications. Linear algebra is also at the foundations of a lot of modern technological fields of study e.g. computer vision, robotics, search engines, machine learning, weather forecasting, data science and so on. It provides a firm foundations for such studies.

The applications of linear algebra are not just restricted to linear systems. Most of the natural dynamical systems have non-linear nature, and therefore are hard to understand completely. The first step in understanding them is to linearize them and understand their linearization. Linear algebra therefore provides us with our first understanding of any natural physical system.

Having so many applications comes with an abstract existence. This is the standpoint we would be taking in this course. We will study vectors from two to n dimensions, systems of linear equations and matrices that provide the bridge between linear equations and vectors, determinants and real linear (vector) spaces. We will learn that matrices can be used to encode transformation of linear spaces, and how to use operations on matrices to determine geometric properties of linear spaces and transformations. Finally we will be able to manipulate such spaces and transformations to solve practical problems that are modelled by linear systems.

Course Aims

This course aims to develop an appreciation of the basic concepts of linear algebra and their applications in geometry and differential equations. This will enable students to address relevant problems in the wider fields of computer science, engineering and natural sciences.

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

1. CLO: Predict whether a given linear system of equations will have any solution(s) and identify the set of solutions.
2. CLO: Explain and evaluate whether a given space is a vector spaces and/or a subspace and find its bases.
3. CLO: Generate a set of orthogonal bases from a given collection of spanning vectors in an abstract space.
4. CLO: Apply the theory of eigenvectors and eigenvalues to solve system of ordinary differential and difference equations
5. CLO: Construct various spaces associated with a linear transformation and formulate their basis-free understanding

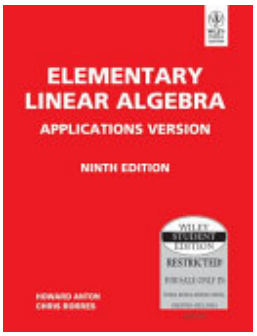
Mode of Instruction

The course content will be delivered mainly through in-class lectures. Students are expected to attend these lectures and understand both the course notes and the book content that is relevant to the course. In the event that we have to shift to Zoom based classes for a length of time, appropriate revisions in assessment and delivery methods will be made.

Engagement & Participation Rules

The plan is for all lectures to be in-person. In case of emergency related measures, the delivery method of synchronous components of the course i.e. lectures, quizzes and office hours will be conducted via Teams. In all of these cases, please login with your Habib ID. Please keep your microphone muted unless you would like to ask a question. This would avoid any background noise for all other participants. We encourage you to also have your camera on, so that your peers can see you. Finally, we expect that you will be respectful to each other always.

Required Texts and Materials



ELEMENTARY LINEAR ALGEBRA APPLICATIONS VERSION, 9TH ED
ISBN: 9788126518692
Edition: 9th Edition
Authors: Howard Anton, Chris Rorres
Publisher: John Wiley & Sons
Publication Date: 2008-01-01

Primary Text

INTERACTIVE LINEAR ALGEBRA
Edition: 1
Authors: Dan Margalit & Joseph Rabinoff
Publisher: Georgia Institute of Technology
Publication Date: 3 June, 2019

Reference Text

Assessments

The following forms of assessments will be used in this course. Grade weight-distribution will be based on the scheme shown in the table below. Each assessment in a single assessment category carries equal weight.

Homework assignments will be set in addition to the below. While there will be no marks gained for these assignments, these will be instrumental in depicting the questions that will appear on the assessments in the table below. So understanding and completing the homeworks regularly will provide an advantage on quizzes and exams. Homework assignment problems will be discussed during the weekly recitation sessions.

No.	Assessment Category	Shorthand	Occurrences	Total Weightage
1	Quizzes	QZ	8 (approx.)	25%
2	Midterm Exams	Mid1/ Mid2	2	40%
3	Group Project	GP	1	5%
4	Final Exam	FE	1	30%

(1) Quizzes: We will be conducting short Quizzes as per the the weekly schedule.
(2) Midterms: The 2 midterms will be held during the weekly recitation sessions, as per the dates mentioned in the weekly schedule.
(3) Final: The final exam will be mandatory and in-person.
(4) Group Project: A project based on applications of Linear Algebra, testing concepts beyond material taught in class. Will be marked based on peer feedback on Canvas.

(5) Student engagement: Student Engagement will be gauged through attendance and quizzes.
You must maintain a **minimum** 90**% on your attendance**/engagement score to sit for the Final Exam.

Here are certain clarifications about the assessment policy in this course. (All quizzes will be in-person, in-class surprise quizzes).

• There will be no make-up homework or assignments that can compensate for the in-class assessments.

• Quizzes can ask questions on anything that has been taught from the beginning of the course to the time
of the quiz. Do not try to split it up by thinking you only need to remain prepared for the stuff taught in
the last one or two weeks. That is a failing strategy.

• The lowest 1 quiz will be dropped from final grade calculations. Do not ask for any further dropped quizzes. Use this wisely since there will be no further concessions on this for any reason. Any further missed quizzes will get an automatic zero.

• In case of an officially granted absence by the RO, any missed quizzes in addition to the above mentioned
dropped one will have the weightage added to the final exam. There will be no other make-ups for these.

• Missing the mid-term exams for any legitimate reason (where legitimacy depends on extreme circumstances like a death in the family or other RO mandated absence from the exam) will add its weightage to the final exam. Missing it for any other reason will result in a straight zero on the exam. There is no make-up midterm exam.

• The Final Exam is comprehensive and mandatory for all students. Failing to appear in the Final Exam will automatically fail you in the course.

Week-Wise Schedule (Tentative)

FALL 2026 Weekly Schedule*

No.	Modules	Weeks	Assessments
1	**Linear Systems**	Week - 0.5 Aug 19 – 21, 2026 Week - 1 Aug 24 – 28, 2026	
2	**Matrices and Determinants**	Week - 2 Aug 31 – Sept 4, 2026 Week - 3 Sept 7 – 11, 2026	**QZ 1**
3	**Vector Spaces and Subspaces**	Week - 4 Sept 14 – 18, 2026 Week - 5 Sept 21 – 25, 2026	**QZ 2 & QZ 3**
4	**Linear (In)dependence, Span and Bases**	Week - 6 Sept 28 – Oct 2, 2026 Week - 7 Oct 5 – 9, 2026 Week - 8 Oct 12 – 16, 2026	**MID 1**
5	**Eigenvalues and Eigenvectors**	Week - 9 Oct 19 – 23, 2026 Week - 10 Oct 26 – 30, 2026	**QZ 4 & QZ 5**
6	**Orthogonality and Gram-Schmidt Process**	Week – 11 Nov 2 – 6, 2026	**QZ 6**

		Week - 12 Nov 9 – 13, 2026 Week - 13 Nov 16 – 20, 2026	**MID 2**
7	**Linear Transformation**	Week - 14 Nov 23 – 27, 2026 Week - 15 Nov 30 – Dec 4, 2026 Week - 16 Dec 7 - 8	**QZ 7 & QZ 8**
Dec 11 - 12, & 14 - 17, 2026			Final Exam

Notes:

* The University reserves the right to correct typographical errors or to adjust the Academic Calendar at any time it deems necessary.

Final Exam Policy

The final exam will be in-person. The Final Exam is comprehensive and mandatory for all students. Failing to appear in the final exam will automatically fail you in the course.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission

of material that has been appropriated,// bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.

b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and

University disciplinary action.

Late Submission Policy

If this is relevant (it is not with any of the assessments in the syllabus currently), there is 10% grade reduction per 24 hours of late submission of assessments.

Academic Integrity & Responsible Use of AI

Generative AI tools such as ChatGPT, Claude, Gemini, DeepSeek, and Copilot may be used in this course as learning and research aids, but they may not replace your own mathematical thinking or contribution.

Permitted uses

Unless an assignment states otherwise, you may use AI to:

- brainstorm ideas, applications, examples, or research directions;
- help you understand a concept or explain something in a different way;
- suggest search terms or potential sources;
- provide feedback on your own writing, organization, or presentation;
- improve grammar, clarity, or language;
- check or troubleshoot work that you have already completed.

You are responsible for independently verifying all mathematical results, factual claims, and sources. AI-generated information is not evidence or a mathematical justification.

Not permitted

You may not:

- have AI complete the mathematical analysis, solution, proof, or other substantive intellectual work that you are being assessed on;
- copy or lightly edit AI-generated text and present it as your own;
- submit AI-generated mathematics, examples, graphs, or explanations that you do not understand and cannot explain yourself;
- use AI to fabricate or supply unverified references, data, or claims;
- use AI to generate your required peer-feedback comments.

Group Project

For the group project, AI may be used for brainstorming, preliminary research, clarification, and feedback on your group's own work. The mathematical analysis, interpretation, verification, and final presentation must be developed and understood by the group. Every group member should be able to explain the mathematics presented in the recording.

Transparency

If you use generative AI in a meaningful way, include a brief AI Use Disclosure with your submission stating the tool(s) used and how they were used. For example:

We used ChatGPT to brainstorm possible applications and to provide feedback on the clarity of an early draft. We independently researched, verified, and developed the mathematical content and final presentation.

You do not need to disclose routine spelling or grammar correction.

Bottom line: Use AI to help you learn, explore, check, and improve—not to do the thinking that this course is designed to assess. If you are unsure whether a particular use is permitted, ask before submitting.

Attendance Policy

The University's standard Attendance Threshold is a minimum 85% of all sessions (i.e., students must be present in at least 85% of all classes/sessions of a given course) scheduled during the course of the semester. However, instructors may nominate a stricter attendance threshold (but cannot relax it more than the aforementioned 85%). Students must abide by these attendance thresholds set by the relevant faculty. If/Once a student's attendance falls within the aforementioned Attendance Threshold in a given course, they cannot receive credit for that course, and will be removed from the course by the OASR. Students must inform their instructor within 48 hours after session, to update the discrepant record on PSCS. So that the same may be corrected manually on the relevant attendance roster by the instructor. In cases where students are absent from a session on account of medical issues or family emergencies, they must inform/intimate the OASR within ten (10) days.

For more information, please refer to Attendance Policy developed by OASR at Habib University.

Program Learning Outcomes (For Administrative Review)

- Problem Analysis

- Design/Development of Solutions
- Investigation

CLOs mapped to Program Learning Outcomes (PLOs)			
CLOs	**PLOs** **PLO1: Problem Analysis** **PLO2: Design/Development of Solutions** **PLO3: Investigation**		
	PLO 1	PLO 2	PLO 3
CLO 1			
CLO 2			
CLO 3			
CLO 4			
CLO 5			

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Mapping of Assessments to CLOs

Mapping of Assessments to CLOs

Assessments	CLO1	CLO2	CLO3	CLO4	CLO5
Q1					
Q2					
Q3					
Q4					
Q5					
Q6					
Q7					
Q8					
Midterm 1					
Midterm 2					
Group Project					

Final					

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.